

The Neuropsychology of Decision-Making: Study & Revision Notes

1. Brain Basics, Neurons, and How the Brain Develops

To understand how we make choices, we first have to understand the physical machine running the show. The brain is built from billions of interconnected cells that communicate using electrical pulses and chemical messengers. It does not arrive fully built at birth; it spends over two decades wiring, trimming, and fine-tuning its connections.

Key Points to Know

- **The Three Main Evolutionary Parts:**

- *Brainstem and Cerebellum:* The oldest, most primitive parts of the brain. They handle automatic survival functions like your heartbeat, breathing, balance, and basic reflexes.
- *Limbic System:* Buried under the cortex, this network manages emotions, stress responses, and survival drives like feeding and reproduction. Its two heavy hitters are the **hippocampus** (vital for forming new memories, spatial navigation, and adult neurogenesis) and the **amygdala** (an almond-shaped structure that tags events with emotion and fear).
- *Cerebral Cortex:* The wrinkled outer "grey matter" layer made of six distinct cellular sheets, divided into two hemispheres by the longitudinal fissure and joined in the middle by the **corpus callosum**. Its bumps are called **gyri** and its grooves are called **sulci**.

- **The Four Cortical Lobes:**

- *Frontal Lobe:* Located at the front, separated from the parietal lobe by the central sulcus. It is the seat of personality and executive functions: planning ahead, impulse control, social rules, and decision-making.
- *Parietal Lobe:* Sits behind the frontal lobe; integrates sensory data from the body like touch, temperature, pressure, pain, and two-point discrimination.
- *Occipital Lobe:* Located at the back of your head; houses the primary visual cortex to process visual features like shape, color, location, and depth.
- *Temporal Lobe:* Located along the sides near the ears; houses the primary auditory cortex (hearing) and handles language recognition and memory storage.

- **Brain Cells (Neurons vs. Glia):**

- *Neurons (~85–100 billion):* The primary information processors. They consist of a **soma** (cell body holding the nucleus), branching **dendrites** (which receive incoming messages), and a long **axon** (which carries electrical impulses away toward other cells).
- *Types of Neurons:* **Sensory neurons** bring environmental inputs inward; **motor neurons** send movement commands to muscles; **pyramidal neurons** are pyramid-shaped excitatory cells heavily involved in advanced thinking and decision-making.
- *Glial Cells:* The non-electrical support staff that outnumber neurons. **Schwann cells** wrap axons in a fatty **myelin sheath** (leaving tiny gaps called *nodes of Ranvier*) to make electrical signals leap faster. **Astrocytes** maintain chemical balance and support new cell growth. **Microglia** act as waste-scavengers and fight inflammation.

- **How Neurons Talk (Synaptic Transmission):**

- Neurons do not physically touch; they are separated by a microscopic fluid gap called the **synaptic cleft**.
- An electrical **action potential** travels down the presynaptic axon to the terminal button.
- This voltage change triggers **synaptic vesicles** to fuse with the membrane (**exocytosis**) and dump chemical messengers (**neurotransmitters**) into the cleft.
- The neurotransmitters cross the gap and bind to matching **receptors** on the postsynaptic dendrite like a key fitting into a lock.

- **Brain Development Timeline:**

- *Prenatal:* Starts 5 weeks after conception; neurons multiply rapidly up to 100 billion, migrate to assigned zones, and begin forming synapses and early myelin.
- *Newborn:* Brain weighs 350–400g. Synapses are strongest in survival and sensory areas (vision and hearing myelinate quickly).
- *Childhood:* The brain hits roughly 90% of its adult volume by age 7. It goes through rapid **synaptogenesis** (building connections) followed by **synaptic pruning** (chopping away unused connections to make networks cleaner and faster).
- *Adolescence:* Total brain volume peaks around age 11–14. Dopamine inputs to the frontal areas surge, while pruning and myelination aggressively streamline circuits.
- *Adulthood:* The prefrontal cortex is the last region to finish maturing (around age 25). Plasticity never stops; our brains form and reshape connections throughout our entire lives.

- **Topographic Organization and How We Study Networks:**

- *Topographic mapping:* Neighboring points in the body or environment connect to neighboring points in the brain, keeping spatial layouts organized and saving wiring energy.
- *Anatomical tract tracing:* Injects fluorescent tracers into animal brains to track physical pathways under a microscope with high precision.
- *ERP / EEG:* Records electrical waves on the human scalp during tasks to time-lock specific choices down to the millisecond.

Everyday Examples

- **The Two-Point Clip Test:** If you unbend a paper clip so it has two points a few millimeters apart, you can easily tell there are two distinct points touching your sensitive fingertip or tongue. But if you press those same two points against your upper arm or back, your parietal cortex only feels a single dull press because sensory receptors there are spaced much farther apart.
- **Phineas Gage:** In 1848, a blasting explosion shot an iron rod through railroad worker Phineas Gage's frontal lobe. He survived, walked, and talked normally, but his personality shifted dramatically: he went from being reliable and polite to impulsive, rude, and incapable of sticking to plans. This historic case proved the frontal lobe controls our personality and social decisions.

Key Takeaways

- Your brain builds most of its 100 billion neurons before birth, but cognitive development is mostly about pruning and insulating (myelinating) those connections later on.
- Communication inside a neuron is electrical (action potential), but communication between neurons is chemical (neurotransmitters across the synapse).
- The cerebral cortex is mapped topographically, meaning physical relationships in the real world are preserved as spatial maps in the brain.
- The prefrontal cortex matures last, taking until about age 25 to finalize its wiring.

2. The Prefrontal Cortex: The Brain's CEO and Decision Engine

The prefrontal cortex (PFC) sits at the very front of the frontal lobe and serves as the brain's executive control center. While older sensory areas just register what is happening right now, your PFC lets you hold goals in mind, imagine things that do not exist yet, weigh future consequences, and choose your actions deliberately.

Key Points to Know

- **What Makes the Human PFC Unique:**
 - It is defined anatomically as the region receiving inputs from the **mediodorsal nucleus of the thalamus**.
 - Evolutionarily, it is the newest and most expanded cortical region in mammals, making up about **29% of the entire human cortex**.
 - Because it takes ~25 years to mature, higher-order abilities like long-term strategic planning and emotional impulse control develop much later than basic memory or movement.
- **Subregions and Their Jobs:**

- *Medial PFC*: Focuses on motivation, drive, and focused attention.
- *Orbital PFC (Orbitofrontal)*: Acts as your impulse control center, keeping risky or inappropriate behaviors in check.
- *Lateral PFC*: Manages working memory, goal-directed planning, and complex multi-step strategies.
- **Cross-Species Comparisons (The Rat Model):**
 - Humans and primates have a specialized "granular" prefrontal cortex that rats physically lack.
 - However, the medial prefrontal cortex of rats shows functional and anatomical homology to primate prefrontal regions. This means studying medial PFC circuits in rats gives us reliable clues about human issues like addiction, depression, and psychosis.
- **Core Theories of PFC Function:**
 - *Integrative Association Hub*: The PFC does not perform one single job; it gathers sensory data, emotions, and past memories from across the whole brain to construct an abstract mental workspace.
 - *Dynamic Filtering Theory*: The PFC acts as a filter, dampening irrelevant distractions and emotional noise so you can focus on the goal at hand.
 - *Miller and Cohen's Integrative Theory*: The PFC builds top-down rules and goals, sending signals back out to other brain regions to steer your thoughts and physical actions.
- **The Anterior-Posterior Connectivity Gradient:**
 - In classic sensory cortex, connections follow clean, matched topographical maps (input and output lines match up in the same spot).
 - Research by Dr. Stacey Bedwell shows that the PFC follows an anatomical gradient: posterior (rear) prefrontal areas are well-aligned like traditional cortex, but as you move toward the anterior (front pole) of the PFC, wiring becomes non-aligned, overlapping, and far more complex.
 - This matches cognitive findings: the anterior pole sits at the very top of the hierarchy, handling the most abstract, open-ended evaluations.
- **Four Everyday Types of Decisions:**
 1. *Routine Choices*: Everyday low-stakes decisions where you have little control over the options (e.g., picking a brand of milk off a grocery shelf).
 2. *Making Outcomes*: Decisions where your personal effort and self-efficacy directly determine success (e.g., planning a project schedule).
 3. *Risky Decisions*: Choices with uncertain, unpredictable outcomes. Adolescents make significantly more risky choices because their PFC wiring is still undergoing heavy pruning and myelination.
 4. *Strategic Decisions*: High-stakes life choices (e.g., changing jobs, buying a house) that require pulling up past knowledge to calculate long-term costs and rewards.

Everyday Examples

- **The Supermarket vs. Project Planning**: When you buy milk, optimism does not change what cartons are on the shelf (routine choice). But when you set out to finish an ambitious design portfolio, your belief in your abilities (self-efficacy) directly improves your focus, execution, and final grade (making outcomes).
- **Filtering Frustration at Work**: If your boss gives an unfair instruction, your immediate emotional impulse might be to shout. An intact PFC applies top-down inhibition, weighing the

consequences of getting fired against the satisfaction of venting, allowing you to stay calm and respond professionally.

Key Takeaways

- The human PFC takes up nearly a third of the cortex and takes roughly 25 years to fully wire up.
- The PFC acts as an association engine, integrating sensory inputs, past memories, and emotional states to run top-down rules.
- Prefrontal anatomy changes across an anterior-to-posterior gradient, with the front-most areas featuring the most complex, non-aligned networks for abstract reasoning.
- Teenagers naturally lean into risky decision-making because their prefrontal inhibitory circuits are still developing.

3. Neurotransmitters, Stress, and Decision Networks

Decisions are not made in a vacuum; they are powered by brain-wide chemical systems and modulated by stress. While amino acid transmitters act like rapid on/off switches, neuromodulator systems project broadly across the cortex to adjust mood, attention, arousal, and value.

Key Points to Know

- **Two Chemical Classes:**
 - *Amino Acids:* The primary local on/off switches. Excitatory transmitters act like an accelerator pedal, while inhibitory transmitters act like a brake to keep brain activity balanced.
 - *Neurotransmitter Systems:* Produced by small groups of cells located subcortically (in the brainstem and midbrain) whose long axons spray outward across the entire cortex to modulate brain states.
- **The Three Big Modulators in Decision-Making:**
 - **Serotonin (5-HT):** Originates in brainstem clusters (like the raphe nuclei in the midbrain, pons, and medulla) and projects to the limbic system, thalamus, and cortex. It regulates mood, sleep, appetite, and the creation of new memories. Low serotonin impairs new memory formation and is tied to depression; SSRIs treat depression by blocking its reuptake back into the presynaptic neuron, leaving more serotonin in the cleft.
 - **Noradrenaline (Norepinephrine):** Produced in the *locus coeruleus* and projects straight to the thalamus, hypothalamus, and PFC. It drives arousal, vigilance, and focused attention. Noradrenergic neurons shut off during REM sleep; disruptions in this system are heavily linked to attention deficits in ADHD.
 - **Dopamine (DA):** Has three major pathways (nigrostriatal, mesolimbic, and mesocortical). It

is the chemical engine of motivation, reward anticipation, novelty seeking, and working memory. Antipsychotics often work as D2-receptor blockers (antagonists) to quiet down overactive dopamine transmission.

- **The HPA Axis and the Stress Response:**

- Formed by a feedback loop between the **Hypothalamus, Pituitary gland, and Adrenal glands**.
- When the amygdala detects a threat, it signals the hypothalamus, activating the HPA axis to release **cortisol** into the bloodstream.
- Cortisol floods the body with glucose for a fight-or-flight response and teams up with adrenaline to forge vivid, high-priority memories (**flashbulb memories**) so you remember how to avoid that danger later.

- **Cognitive Decision Models:**

- *Epstein's Cognitive-Experiential Self-Theory*: We have two parallel decision engines. The **rational system** is slow, analytical, conscious, effortful, and relies on the PFC and hippocampus. The **experiential system** is fast, automatic, emotional, intuitive, and driven by the amygdala and striatum.
- *Shallice's Supervisory System*: Everyday routine tasks run on **contention scheduling** (unconscious, learned action scripts called *schemas*). When you encounter a completely new or conflicting problem, the **Supervisory Attentional System (SAS)** intervenes to deliberately steer behavior.
- *Stuss & Benson's Hierarchy*: A three-tier pyramid: automatic sensory routines at the bottom, goal-directed executive control in the middle, and **metacognition** (self-reflection on your own thought processes) at the very top.

- **The Phone-Ringing Processing Cascade:**

- Level 1 (Sensory/Premotor): You hear a generic notification ping and automatically pick up your phone.
- Level 2 (Contextual): You hear the ringtone, but you are driving or in a meeting with your boss, so you suppress the reach.
- Level 3 (Episodic/Prior Knowledge): You remember a family member went into surgery this morning, so you override the meeting context and check it anyway.
- Level 4 (Anterior PFC Branching): You balance multiple sub-goals and open-ended outcomes to decide whether checking now or following up later fits your larger personal priorities.

Everyday Examples

- **The Red Traffic Light**: When a red light flashes, you do not sit and ponder the physics of braking; your brain executes a practiced schema via contention scheduling—foot off the gas, onto the brake. But if an ambulance suddenly speeds the wrong way through that intersection, your Supervisory Attentional System (SAS) must step in instantly to override the green light and hit the brakes.
- **Flashbulb Stress Memories**: If you are attacked by an aggressive dog on a specific jogging trail, your HPA axis floods your brain with cortisol. That surge burns the sensory details of the trail into your memory, ensuring that the next time you approach that corner, your rational and experiential systems steer you down a different path.

Key Takeaways

- Local amino acids provide fast on/off signaling, while subcortical systems (dopamine,

serotonin, noradrenaline) broadcast mood, attention, and reward states brain-wide.

- Epstein's model splits thinking into a fast, emotional "experiential" system and a deliberate, logical "rational" system.
- Frontal lobe damage leaves automatic schemas intact (contention scheduling works) but breaks your ability to adapt to brand-new, confusing situations (SAS fails).
- Decision control is organized along an anterior-to-posterior hierarchy, running from simple physical actions at the back to abstract context and self-reflection at the front.

4. Decision-Making Styles and Childhood Influences

People approach choices using characteristic habits known as decision-making styles. These styles are not hardcoded at birth; they are shaped throughout childhood by attachment, environmental stability, and social interactions.

Key Points to Know

- **Defining Decision-Making Style:** A habitual, learned pattern a person uses when gathering information, weighing options, and making choices.
- **Early Theoretical Milestones:**
 - *McKinney & Keene (1974)*: Split thinkers into systematic (structured, logical step-followers) versus intuitive (exploratory, trial-and-error learners).
 - *Driver (1979)*: Proposed that styles are learned behaviors based on how much data someone collects and how many alternatives they review, suggesting people have primary and secondary styles.
 - *Harren (1979)*: Established three core approaches: **Rational** (takes personal ownership, plans logically), **Intuitive** (takes personal ownership, follows internal gut feelings), and **Dependent** (defers ownership onto other people).
- **Scott & Bruce's Five GDMS Styles (General Decision Making Style):**
 1. *Rational*: Methodical, slow, gathers all facts, and systematically weighs alternatives.
 2. *Intuitive*: Relies heavily on gut hunches, feelings, and what "feels right" in the moment.
 3. *Dependent*: Uncomfortable deciding alone; constantly seeks advice, validation, and direction from others.
 4. *Avoidant*: Feels anxious and uneasy about making choices; procrastinates and only decides when forced by deadlines.
 5. *Spontaneous*: Impulsive and fast; makes snap choices just to get through the process without overthinking.
- **GDMS Research Findings (Bavol'ar & Orosova, 2015):**
 - Rational and Dependent styles correlate positively (seeking expert advice is often part of a thorough, rational information-gathering strategy).
 - Rational and Avoidant styles correlate negatively (people who plan ahead do not avoid problems).

- Avoidant decision-making correlates significantly with poorer mental health and higher psychological distress.
- **How Childhood Trauma Alters the Machinery:**
 - The PFC has an extended developmental window, making its developing synapses vulnerable to severe stressors.
 - Repeated childhood trauma creates a "dose-response" effect: frequent trauma chronically dysregulates the HPA axis, flooding the system with cortisol and reducing synaptic plasticity.
 - Early trauma impairs executive functions and working memory, often pushing adults toward extreme avoidant or purely intuitive styles.
- **The Surprising Truth About Sibling Aggression (Dr. Bedwell's Lab):**
 - Sibling conflict is common, but using aggression to establish dominance and hoard possessions during childhood alters prefrontal development.
 - Counterintuitively, it is the *perpetrators* who used aggression to maintain dominance who are significantly more likely to display an **avoidant decision-making style** as adults, driven by underlying social anxiety and fear of failure.
 - Furthermore, while general trauma increases real-time reckless behavior, sibling aggression does *not* predict risky decision-making on laboratory tasks, showing that everyday family conflict affects decision styles differently than major trauma.

Everyday Examples

- **Buying a Car Across the Five Styles:**
 - The *rational* buyer builds a spreadsheet comparing fuel economy and safety ratings.
 - The *intuitive* buyer sits in the driver's seat and buys it because it "feels right".
 - The *dependent* buyer brings three family members along to choose for them.
 - The *avoidant* buyer leaves the broken car in the driveway for months rather than face the dealership.
 - The *spontaneous* buyer signs the paperwork on the first car parked by the door just to get it over with.
- **Childhood Dominance Leading to Adult Avoidance:** A kid who constantly shoves their younger sibling to keep total control of toys grows up with fragile, dominance-based coping habits. As an adult in a workplace where physical bullying is not an option, they lack collaborative negotiation skills and retreat into an avoidant style whenever tough choices arise.

Key Takeaways

- Decision styles are learned behavioral habits, and most people blend styles depending on the situation.
- Chronic avoidance of decisions is strongly linked to higher anxiety and worse mental health outcomes.
- Severe early trauma alters the physical development of the HPA axis and prefrontal cortex, locking people into rigid cognitive patterns.
- Childhood sibling aggression specifically predicts adult avoidance (in those driven by dominance), but it does not drive risky gambling choices.

5. Decision Deficits and Clinical Conditions

A decision-making deficit occurs when brain circuits are disrupted, leaving an individual unable to adapt to changing rules, weigh rewards against punishments, or resist impulsive actions.

Comparing these breakdowns across clinical conditions shows how fragile the prefrontal network really is.

Key Points to Know

- **Substance Use Disorder (Stimulants):**
 - Chronic cocaine or methamphetamine abuse dumps massive amounts of dopamine into the striatum and prefrontal circuits.
 - This accelerates the transition from flexible, goal-directed behavior into rigid, automated **habits**.
 - *Reversal Learning Deficit:* Patients become "stuck"; even when rules change and a behavior causes immediate harm, their brains fail to update, keeping them locked in compulsive drug-seeking.
- **Schizophrenia (A Disorder of Connectivity):**
 - Characterized by positive symptoms (delusions, hallucinations) and negative symptoms (blunted affect, withdrawal).
 - Structural scans show reduced prefrontal white matter volume, altered cellular organization, and smaller frontal lobes.
 - Excessive dopamine in the striatum impairs how the brain integrates cortical information, causing patients to "jump to conclusions" based on very little data and persist with bad choices.
- **Autism Spectrum Conditions:**
 - Adults with autism report that decision-making is mentally exhausting, overwhelming, and anxiety-provoking, often leading to decision avoidance.
 - Quick decisions or sudden disruptions to routines are especially difficult.
 - On experimental tasks, they often rely heavily on a rigid **win-stay-lose-shift** strategy, working harder for rewards while showing reduced internal sensitivity to reward feedback.
- **Major Depressive Disorder:**
 - Characterized by blunted striatal and prefrontal responses to positive rewards, along with low serotonin transmission in the PFC.
 - **Anhedonia** (inability to feel pleasure) is directly tied to frontostriatal reward circuit breakdown.
 - Depleted prefrontal serotonin leads to rigid, inflexible thinking and poor problem-solving—a

- deficit that is significantly worse in individuals with a history of suicide attempts.
- *Depressive Realism (Alloy & Abramson)*: The controversial finding that people with mild depression sometimes evaluate their lack of control over the world more accurately than non-depressed people (who hold an optimistic self-serving bias).
- *Analytical Rumination Hypothesis*: The idea that depressive rumination originally evolved to help people isolate themselves and thoroughly analyze complex social dilemmas.
- **Psychopathy / Antisocial Personality Disorder**:
 - Marked by shallow emotional affect, callousness, impulsivity, and grandiosity.
 - Driven by a functional disconnect between the **amygdala** (which processes fear and punishment) and the **prefrontal cortex** (which anticipates future consequences).
 - Because they do not feel the anticipatory emotional warning signal of punishment, they make reckless choices focused purely on immediate gratification.
- **Laboratory Tests of Decision-Making**:
 - **Iowa Gambling Task (IGT)**: Simulates real-world risk and reward. Players choose from four card decks. Decks A and B offer big immediate cash payouts but hidden catastrophic penalties (disadvantageous). Decks C and D offer small payouts but tiny penalties, resulting in long-term profit (advantageous). Healthy people quickly sense the risk and stick to C and D; patients with frontal lobe or amygdala damage keep gambling on A and B until they go broke.
 - **Eriksen Flanker Task**: Tests cognitive control and response inhibition. You must identify a central target arrow while being flanked by distracting arrows that either match (congruent: >>>>) or clash (incongruent: >><>). Resolving the clash requires executive prefrontal control.
- **Rehabilitation**: While medication rarely restores damaged decision-making logic on its own, behavioral therapies like **Cognitive Remediation Therapy (CRT)** train the brain through targeted mental exercises to adopt alternative problem-solving strategies.

Everyday Examples

- **The Broken Traffic Light Switch**: Imagine the city quietly flips traffic rules overnight so green means stop and red means go. A neurotypical driver gets confused, slams the brake at a green light once, and rapidly adapts their habits. A patient with stimulant dependency or schizophrenia keeps driving through green lights and crashing, because their frontostriatal circuits cannot update the old rule.
- **The Flanker Distraction**: When you try to hit the "Submit" button on a messy website covered in flashing banner ads pointing the other way, you hesitate for a split second. That tiny delay is your prefrontal cortex working hard to suppress the distracting noise and keep your motor response on target.

Key Takeaways

- Stimulant abuse turns flexible, goal-directed planning into rigid, unbending motor habits.
- Schizophrenia is primarily a neural connectivity disorder, causing patients to jump to conclusions with minimal evidence.
- Adults with autism often find rapid choices exhausting, which can lead to decision avoidance; clear parameters and extra time help relieve this strain.
- On the Iowa Gambling Task, frontal lobe damage prevents people from learning from past penalties, keeping them trapped in risky choices.

Glossary

- **Action Potential:** A rapid electrical impulse that travels down an axon to deliver a signal to the end of a neuron.
- **Amygdala:** An almond-shaped limbic structure responsible for processing fear, emotional tags, and threat signals.
- **Anhedonia:** The inability to experience pleasure or reward from everyday activities; a hallmark of depression.
- **Astrocytes:** Star-shaped glial cells that maintain the brain's chemical balance, protect tissue, and support neurogenesis.
- **Autonomic Schemas (Contention Scheduling):** Well-rehearsed, unconscious action routines triggered automatically by familiar cues.
- **Cognitive Remediation Therapy (CRT):** A behavioral training approach that exercises brain networks to improve memory, planning, and decision strategies.
- **Corpus Callosum:** The thick nerve tract connecting the left and right cerebral hemispheres.
- **Depressive Realism:** The phenomenon where individuals with mild depression make more accurate judgments about their lack of control than overly optimistic peers.
- **Dopamine:** A neurotransmitter critical for anticipating rewards, driving motivation, sustaining working memory, and executing motor choices.
- **Eriksen Flanker Task:** An experiment measuring cognitive control and response inhibition by testing how well someone ignores distracting, conflicting visual cues.
- **General Decision Making Style (GDMS):** A 5-factor psychological questionnaire measuring whether someone is Rational, Intuitive, Dependent, Avoidant, or Spontaneous.
- **HPA Axis:** The Hypothalamic-Pituitary-Adrenal network that manages physical stress responses and pumps out cortisol.
- **Iowa Gambling Task (IGT):** A card game assessment that evaluates real-world decision-making under risk and ambiguity by tracking how players navigate good vs. bad decks.
- **Microglia:** Dynamic glial scavenger cells that clear cellular debris and manage inflammation in the nervous system.
- **Myelin Sheath:** The protective fatty wrapping around axons produced by glial cells that allows electrical signals to travel significantly faster.
- **Prefrontal Cortex (PFC):** The most anterior region of the frontal lobe; the brain's executive control center for planning, impulse regulation, and complex decisions.
- **Pyramidal Neurons:** The primary excitatory brain cells in the cortex, easily recognized by their triangular cell bodies.
- **Response Inhibition:** The executive ability to deliberately suppress actions or impulses that are inappropriate or interfere with a long-term goal.
- **Serotonin:** A neuromodulator that regulates mood, sleep, appetite, and memory; low levels are linked to depression and cognitive rigidity.
- **Supervisory Attentional System (SAS):** Shallice's term for the conscious, top-down executive system that steps in to guide decisions during brand-new or conflicting situations.
- **Synaptic Pruning:** The developmental process of eliminating unused synapses to make neural communication more efficient.

- **Synaptogenesis:** The rapid creation of new synaptic junctions between neurons.
- **Topographic Organization:** The spatial arrangement where adjacent points in sensory inputs or the body are mapped onto adjacent neurons in the cortex.

Quick Revision Sheet

Core Ideas

- **Development Lag:** The brain reaches full physical size in early adolescence (11–14), but the prefrontal cortex continues pruning and myelinating until roughly age 25.
- **Integrated Control:** The prefrontal cortex acts as an association hub. It does not work alone; it integrates emotional inputs from the amygdala and memory from the hippocampus to

direct thoughts and behavior.

- **Dual Engines:** Decisions balance fast, emotional, automatic systems (experiential / contention scheduling) with slow, deliberate, top-down systems (rational / SAS).
- **Experiences Build Biology:** Childhood environments physically alter development. Chronic stress dysregulates the HPA axis, while early dominance aggression with siblings alters prefrontal development, leading to an avoidant decision style in adulthood.

Core Theories and Models at a Glance

Framework	Theorist	How It Explains Thinking & Choosing
Cognitive-Experiential Theory	Epstein	We balance two systems: an emotional, automatic, effortless Experiential system (amygdala/striatum) and an analytical, slow, effortful Rational system (PFC/hippocampus).
Supervisory Attention Model	Shallice	Routine habits run on automatic Contention Scheduling ; novel, ambiguous, or dangerous situations force the conscious Supervisory Attentional System (SAS) to take over.
Hierarchical Metacognition	Stuss & Benson	A three-tier pyramid: bottom = automatic sensory processing; middle = executive goal direction; top = metacognition (reflecting on your own choices).
Integrative Theory of PFC	Miller & Cohen	The PFC acts as a rule-maker, using top-down guidance to steer lower sensory and motor systems toward overarching goals.
GDMS Model	Scott & Bruce	People rely on 5 learned decision styles: Rational, Intuitive, Dependent,

		Avoidant, and Spontaneous.
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Decision Tasks & Clinical Breakdown

Assessment / Condition	What It Reveals
Iowa Gambling Task (IGT)	Tests ability to learn from risk and penalty. Patients with frontal lobe damage or psychopathy fail to realize decks A & B lead to net ruin.
Eriksen Flanker Task	Tests response inhibition and cognitive control under distraction (>>>> vs ><>>).
Substance Use Disorder	Stimulants flood dopamine pathways, causing people to transition from flexible planning to rigid, unbreakable motor habits (reversal learning failure).
Schizophrenia	Disordered prefrontal-striatal connectivity leads to jumping to conclusions based on minimal facts.
Autism	Decision-making is experienced as exhausting and anxiety-provoking, often showing strong win-stay-lose-shift rigidity.
Depression	Reduced striatal reward sensitivity and blunted prefrontal serotonin cause anhedonia and cognitive rigidity.
Psychopathy	Amygdala-PFC disconnection eliminates the anticipatory fear of punishment, prompting risky, destructive choices.

Most Critical Distinctions

- **Action Potential vs. Synaptic Transmission:** Electrical signal running *inside* one cell vs. chemical molecules floating *between* two cells.
- **Synaptogenesis vs. Pruning:** Building millions of new connections vs. cutting away the weak, unused ones to increase speed and efficiency.
- **Routine Choices vs. Making Outcomes:** Picking fixed options where you have zero control (buying milk) vs. situations where your confidence and effort change the outcome (running a project).
- **Sibling Aggression vs. General Trauma:** Severe trauma triggers reckless, impulsive gambling; sibling dominance aggression specifically builds adult avoidance habits without

increasing real-time gambling risk.